SOFTWARE REQUIREMENTS SPECIFICATION

EXPLORATORY DATA ANALYSIS OF RETAIL TRANSACTIONS

Exploratory Data Analysis using Python

DOCUMENT CONTROL	DETAILS
Document type	Software Requirements Specification (SRS)
Version	1.0
Prepared for	Academic Data Analytics Project
Project title	Exploratory Data Analysis of Retail Transactions
Dataset	Retail_Transaction_Dataset.csv
Status	Initial requirements baseline

1. Introduction

1.1 Purpose

This SRS specifies the requirements for a data analytics project that explores retail transaction records to identify purchasing patterns, product and category performance, payment preferences, time-based trends, and the relationship between discounts and transaction value. It serves as the baseline for implementation, testing, documentation, and evaluation.

1.2 Project Scope

The project will load, validate, clean, transform, analyse, and visualise the provided retail transaction dataset. It will use descriptive statistics and exploratory data analysis (EDA) to produce evidence-supported findings and practical recommendations. The deliverable is an explanatory, reproducible notebook and a project report; it is not a production point-of-sale system, inventory platform, or real-time prediction service.

1.3 Intended Users

- Student/project developer — implements the analysis and documents decisions.
- Project guide/evaluator — checks requirements, analytical quality, and evidence.
- Retail/business reader — uses the findings to understand sales patterns and potential actions.

1.4 Definitions and Acronyms

Term	Meaning
EDA	Exploratory Data Analysis: examining data using summaries and visualisations.
SRS	Software Requirements Specification.
Data quality	The completeness, validity, consistency, and uniqueness of the records.
Feature engineering	Creating useful analytical fields from existing columns, such as month or day of week.
KPI	Key Performance Indicator, such as total sales or average transaction amount.

1.5 References and Source Notes

The analysis will use the supplied Retail_Transaction_Dataset.csv file. The original download/source URL was not included in the information available when this SRS was prepared; the final report must record the actual source URL or provider, access date, and any licence/attribution requirements. Do not invent a source.

Recommended technical references: official documentation for Python, pandas, NumPy, Matplotlib, Seaborn, and Jupyter/Google Colab.

2. Overall Description

2.1 Product Perspective

This is a standalone, notebook-based analytical workflow. The notebook will contain code cells, Markdown explanations, tables, charts, findings, and recommendations. The analysis should be rerunnable from the raw CSV with clearly documented processing steps.

2.2 Dataset Overview

A preliminary inspection of the supplied CSV found 100,000 rows and 10 columns. The inspected file contains no null values in the ten columns. These observations are initial checks only; duplicates, data types, value ranges, category consistency, date parsing, and formula consistency must still be validated during cleaning.

Attribute	Expected meaning / use
CustomerID	Customer identifier; use for distinct-customer counts and repeat-customer exploration, not as a continuous numeric measure.
ProductID	Product identifier/category code such as A, B, or C.
Quantity	Number of items in the transaction record.
Price	Unit price in USD, as described in the dataset metadata.
TransactionDate	Transaction date and time; parse into a datetime type.
PaymentMethod	Payment method category.
StoreLocation	Store address/location text; consider grouping cautiously because many unique values may exist.
ProductCategory	Product category such as Home Decor, Books, or Clothing.
DiscountApplied(%)	Discount percentage applied to the transaction.
TotalAmount	Total paid after discount, according to the dataset description.

2.3 Assumptions and Dependencies

- The supplied CSV is the raw input and will be preserved unchanged.
- Python, pandas, NumPy, Matplotlib, and Seaborn will be used; Plotly is optional.
- Jupyter Notebook or Google Colab will be available.
- Currency values are described as USD; the project will not convert currencies.
- The exact business meaning of each row (individual item line versus whole transaction) must be confirmed from the dataset source. Until confirmed, metrics will be described as record-level transaction measures where appropriate.
- The relationship used to calculate TotalAmount should be checked against Quantity, Price, and DiscountApplied(%); discrepancies must be reported rather than silently assumed.

2.4 Constraints

- The dataset requirement is at least 500 records and 10 features; the inspected CSV currently meets the numeric threshold with 100,000 rows and 10 columns.
- The notebook must include at least 10 distinct, clearly labelled visualisations.
- The project must present at least 5 evidence-supported insights.
- The final submission must include the raw dataset, completed notebook (.ipynb), and report/documentation.
- No claims about profit, inventory levels, marketing campaign causality, or customer demographics may be made unless the data contains the required fields.

3. Problem Statement

Retail transaction data can contain useful signals about product demand, customer purchasing activity, payment choices, sales timing, and discount usage. Without systematic analysis, these patterns may be difficult to identify and communicate. This project will apply data cleaning, descriptive statistics, and EDA to the available transaction records and present findings that can support informed retail decisions, while clearly stating the limitations of the dataset.

4. Project Objectives

- Validate whether the dataset is suitable for a retail analytics assignment.

- Assess missing values, duplicate records, invalid values, inconsistent categories, and data types.
- Prepare analysis-ready date/time, numeric, and categorical fields.
- Summarise transactions, quantities, prices, discounts, and total amounts with descriptive statistics.
- Conduct univariate, bivariate, and multivariate EDA.
- Compare transaction amounts and quantities by product, category, payment method, time period, and available location fields.
- Examine correlations and relationships among suitable numeric variables without implying causation.
- Create at least 10 useful visualisations and derive at least 5 findings supported by tables, metrics, or charts.
- Provide actionable recommendations, limitations, future scope, and a reproducible final notebook.

5. Functional Requirements

ID	Requirement	Description	Priority
FR-01	Input loading	The notebook shall load the CSV using pandas and display the dataset shape, column names, sample records, and data types.	Required
FR-02	Initial profiling	The notebook shall calculate descriptive summaries and report null counts, duplicate-row counts, unique counts, and basic numeric ranges.	Required
FR-03	Data validation	The notebook shall validate date parsing, positive/non-negative Quantity and Price rules, discount range, non-negative TotalAmount, and category consistency.	Required
FR-04	Data cleaning	The notebook shall document any missing-value, duplicate, invalid-value, and inconsistent-category treatment. No record shall be removed without a documented reason.	Required
FR-05	Feature engineering	The notebook shall derive relevant date features such as year, month, month name, day of week, and hour when valid dates are available.	Required
FR-06	Descriptive analysis	The notebook shall calculate relevant summary metrics, including record count, distinct customers/products, quantity totals, total amount, average amount, and discount summaries.	Required
FR-07	Univariate EDA	The notebook shall analyse distributions of numeric variables and frequencies of categorical variables.	Required
FR-08	Bivariate EDA	The notebook shall compare relevant variable pairs, such as category versus amount and payment method versus amount.	Required
FR-09	Multivariate EDA	The notebook shall investigate patterns across multiple variables, such as category, time period, and total amount.	Required
FR-10	Groupby and pivot analysis	The notebook shall use groupby summaries and pivot tables for comparisons across product/category, payment method, and time.	Required
FR-11	Correlation analysis	The notebook shall compute and visualise correlations for appropriate numeric variables and explain that correlation does not establish causation.	Required
FR-12	Visualisation	The notebook shall include at least 10 distinct charts with meaningful titles, axis labels, legends where needed, and readable formatting.	Required
FR-13	Insights	The notebook/report shall present at least 5 findings, each linked to quantitative evidence or a visualisation.	Required
FR-14	Recommendations	The final report shall provide recommendations that follow from the evidence and acknowledge analytical limitations.	Required
FR-15	Reproducibility	The notebook shall run from top to bottom with explanatory Markdown and clearly commented code where useful.	Required
FR-16	Final deliverables	The project shall retain the original CSV and submit the completed .ipynb notebook and final report/SRS as required by the project guide.	Required

6. Non-Functional Requirements

ID	Requirement
NFR-01 Usability	Markdown cells shall explain the purpose and interpretation of each major analysis section.
NFR-02 Accuracy	Calculations shall use appropriate data types and documented formulas; totals and derived values shall be sanity-checked.

NFR-03 Reproducibility	The notebook shall use a clear execution order and avoid dependence on undocumented manual steps.
NFR-04 Maintainability	Code shall use descriptive variable names, reusable functions where appropriate, and comments for non-obvious logic.
NFR-05 Performance	The workflow shall handle the current 100,000-row dataset on a typical student computer or hosted notebook without unnecessary repeated processing.
NFR-06 Visual clarity	Charts shall include informative titles, labelled axes, legible text, and legends where applicable.
NFR-07 Traceability	Cleaning decisions, assumptions, metrics, and findings shall be documented so that an evaluator can trace conclusions to data.
NFR-08 Privacy and responsible use	CustomerID and StoreLocation shall be used only as needed for aggregate analysis; individual-level information shall not be exposed in the report.

7. Data Processing and Analysis Requirements

7.1 Data Quality Checks

- Check the number of rows/columns, column names, data types, and duplicate rows.
- Check missing values, whitespace, inconsistent category labels, and unexpected category values.
- Parse TransactionDate using a documented date format and identify any failed conversions.
- Check Quantity and Price for values outside the expected domain; investigate rather than automatically delete unusual records.
- Check DiscountApplied(%) against the expected percentage range (normally 0–100) and investigate any out-of-range values.
- Check TotalAmount for negative values and compare it with a candidate calculation such as Quantity × Price × (1 − discount/100), while confirming the source's exact calculation rules.
- Retain a record of the number of rows before and after each cleaning step.

7.2 Feature Engineering

- Create Year, Month, MonthName, DayOfWeek, and Hour from TransactionDate.
- Create derived measures only where definitions are defensible, for example GrossAmount = Quantity × Price.
- If a calculated discount amount is used, document the formula and verify its compatibility with TotalAmount.
- Avoid treating CustomerID as a continuous numeric variable in correlation or distribution analyses.

7.3 Required Analysis Areas

Analysis type	Examples
Univariate	Distribution of TotalAmount, Quantity, Price, DiscountApplied(%); frequency of ProductCategory and PaymentMethod.
Bivariate	TotalAmount by ProductCategory; Quantity by ProductID; discount versus TotalAmount; payment method versus transaction amount.
Multivariate	Monthly total amount by category; category and payment-method comparisons; time-of-day patterns across categories.
Groupby / Pivot	Count, sum, mean, median, and quantity summaries grouped by relevant dimensions.
Correlation	Correlation matrix for meaningful numeric measures such as Quantity, Price, discount, and TotalAmount, with caveats about formula-induced relationships.

7.4 Minimum Visualisation Plan

No.	Visualisation	Purpose
1	Histogram of TotalAmount	Distribution and skewness of transaction amounts.
2	Box plot of TotalAmount	Spread and potential outliers.
3	Bar chart of ProductCategory counts	Most frequent categories in the records.
4	Bar chart of PaymentMethod counts	Observed payment-method frequency.
5	Bar chart of total amount by ProductCategory	Category-level total amount.
6	Bar chart of quantity by ProductID	Product-code quantity comparison.
7	Line chart of monthly total amount	Time-based trend, if date coverage supports it.
8	Bar chart of average TotalAmount by PaymentMethod	Average amount comparison by payment method.
9	Scatter plot of discount versus TotalAmount	Relationship between discount and paid amount.
10	Correlation heatmap	Relationships among selected numeric variables.
11	Box plot of TotalAmount by ProductCategory	Compare spread across categories.
12	Bar/line chart of transactions by day of week or hour	Temporal activity patterns.

The final notebook must contain at least 10 distinct visualisations. This list is a plan; chart selection should be adapted if a variable has too many unique values or if data validation shows a chart would be misleading.

8. Output Requirements

- A completed Jupyter Notebook (.ipynb) containing code, Markdown explanations, tables, charts, and findings.
- The original raw CSV preserved as the source dataset.
- A cleaned analysis dataset or clearly documented in-notebook cleaning result, where appropriate.
- A final report containing introduction, problem statement, objectives, dataset description and source, cleaning/preprocessing, analysis, visualisations, at least 5 insights, recommendations, conclusion, limitations, future scope, and references.
- This SRS document as the initial requirements specification.

9. Acceptance Criteria and Testing

ID	Acceptance criterion
AC-01	The CSV loads successfully and the notebook reports the correct shape and columns.
AC-02	The dataset contains at least 500 rows and 10 features; initial inspection shows 100,000 rows and 10 columns.
AC-03	Nulls, duplicates, invalid ranges, date parsing, and category consistency are checked and documented.
AC-04	Every cleaning change is explained and row counts before/after cleaning are reported.
AC-05	Univariate, bivariate, and multivariate analyses are included, along with groupby/pivot summaries and correlation analysis.
AC-06	At least 10 distinct visualisations are rendered and include appropriate titles and labels.
AC-07	At least 5 insights are supported by specific values, tables, or charts rather than unsupported opinion.
AC-08	Recommendations are linked to findings and do not claim causation or profitability without supporting data.
AC-09	The notebook executes from top to bottom without unresolved errors in the intended environment.
AC-10	The final report identifies the actual dataset source and documents limitations and future scope.

10. Project Limitations

- The listed columns do not include cost of goods sold or operating costs; therefore, profit and margin cannot be calculated reliably.

- The dataset description lists ProductID values such as A, B, and C, which may not uniquely identify real-world products beyond those codes.
- StoreLocation may contain many unique addresses, limiting direct visualisation without appropriate aggregation or geocoding.
- CustomerID supports repeat-purchase counts, but the dataset does not provide customer demographics or explicit loyalty-program information.
- The data alone cannot establish that discounts caused changes in purchase behaviour.
- The source, sampling method, and whether each row represents an entire transaction or a line item must be verified and documented.

11. Future Scope

- Add inventory, product cost, returns, and margin data to support stock and profitability analysis.
- Integrate reliable store metadata and geographic coordinates for regional comparisons, where privacy and quality allow.
- Use a longer and well-documented time series for seasonal analysis and forecasting.
- Build an interactive dashboard using Plotly or a dashboard framework after validating the core analysis.
- Develop customer segmentation or forecasting models only if the dataset quality, sample design, and project scope justify them.

12. Conclusion

The supplied retail dataset is suitable for a structured exploratory retail analytics project at the dataset-size and feature-count level. The preliminary inspection found 100,000 records, 10 columns, and no null values in the inspected columns. Final suitability depends on further checks for duplicates, valid ranges, date consistency, amount calculations, and the source's row-level meaning. Meeting the requirements in this SRS will produce a reproducible notebook with documented data preparation, at least 10 visualisations, evidence-supported insights, recommendations, and transparent limitations.

— End of Specification —

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